

Build Your First AI Prototype

How to use AI and your design system to prototype smarter, not screen by screen

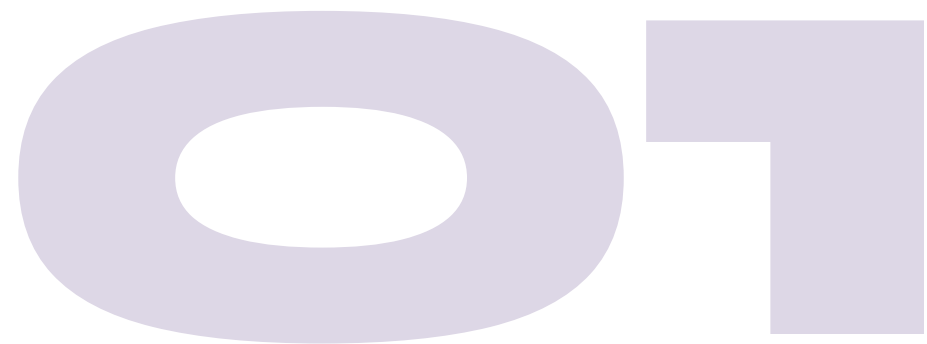


Winnie
Nguyen

Senior Product Designer at NAB

Former Sr Product Designer at Go1

Former Product Design Instructor at StayLab



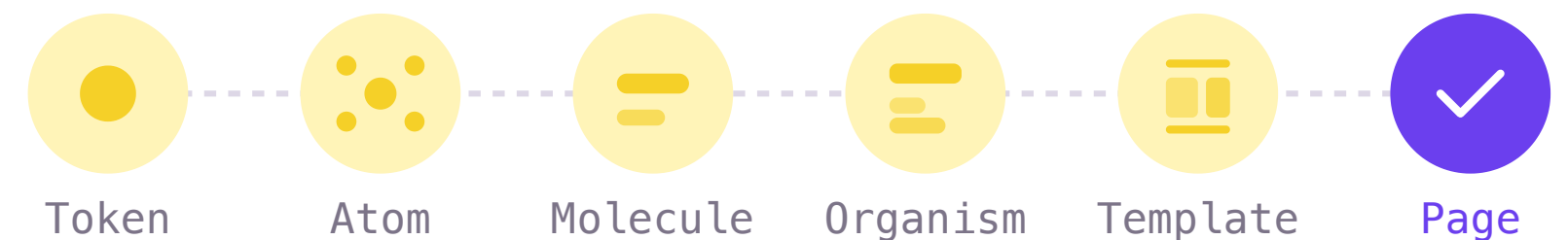
Where we left off

BUILDING ON YOUR DESIGN SYSTEM FROM LAST SESSION

— LOOK WHAT YOU MADE

Last session, you built something **real.**

That's a complete design system, and you built it.
Most people skip this step and start prototyping
from nothing. You've got a head start.

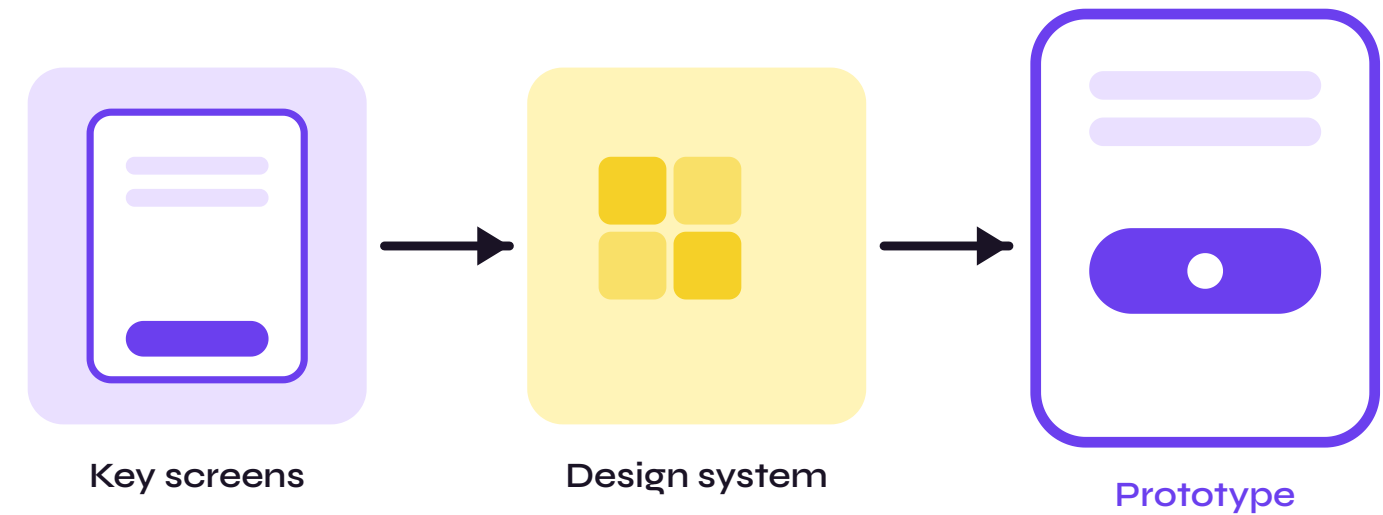


All done. All yours.

— SO... WHAT NOW?

From your screens, to a system, to something that runs.

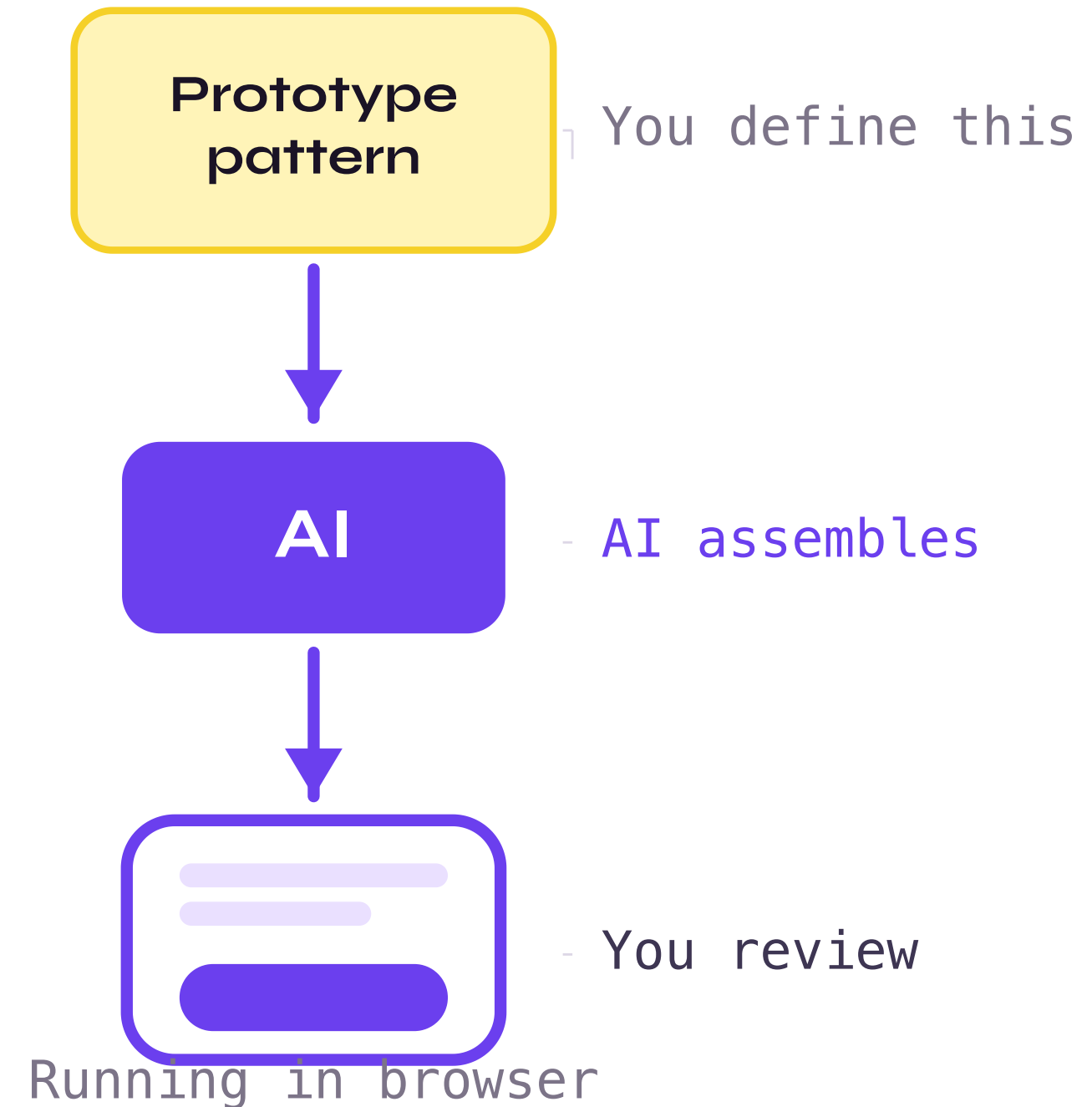
Your key screens already hold the system. Pull it out, hand it to AI, and get back a prototype that moves.



— LET'S BE SPECIFIC

Not a Figma file. Not a mockup. A prototype that actually runs.

Real code running in a browser, not a picture of an app. You define the system once; AI assembles the screens from it.



— WHY THIS SESSION EXISTS

Traditional prototype vs AI-assisted prototype.

TRADITIONAL PROTOTYPE

Every screen by hand.

You design each screen manually. Every state, every interaction, every component placed one at a time.

Changes require updating every screen individually.

AI can't build from it. There's no single source of truth.

Scales to 3 screens. Falls apart at 10.

AI-ASSISTED PROTOTYPE

Define once. Build many.

You define the prototype pattern once: what components exist (with states) and how screens connect.

AI reads the prototype pattern and generates every screen from it.

Changes update from the pattern, not from each individual screen.

Scales to 30 screens as easily as 3.

02

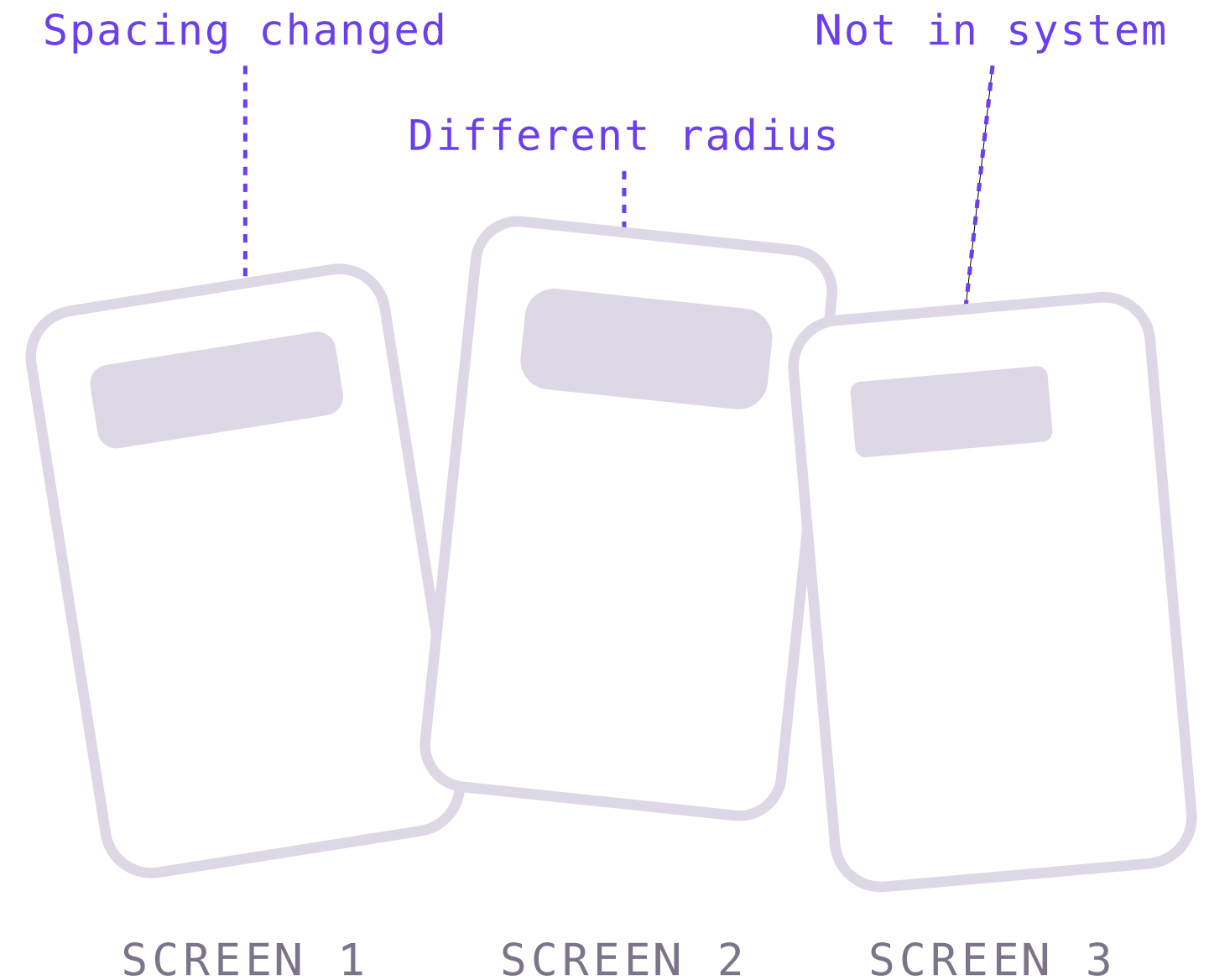
The problem with screen-by-screen

WHY STARTING WITH A SCREEN IS THE WRONG MOVE

— THE MOST COMMON MISTAKE

Screen by screen. Everything drifts.

Home, then detail, then settings. By screen three the button has drifted and the spacing has moved. Hand that to AI and every prompt starts from zero, because nothing connects the screens.



o3

Get everything ready before hand

THE PATTERN-FIRST SOLUTION

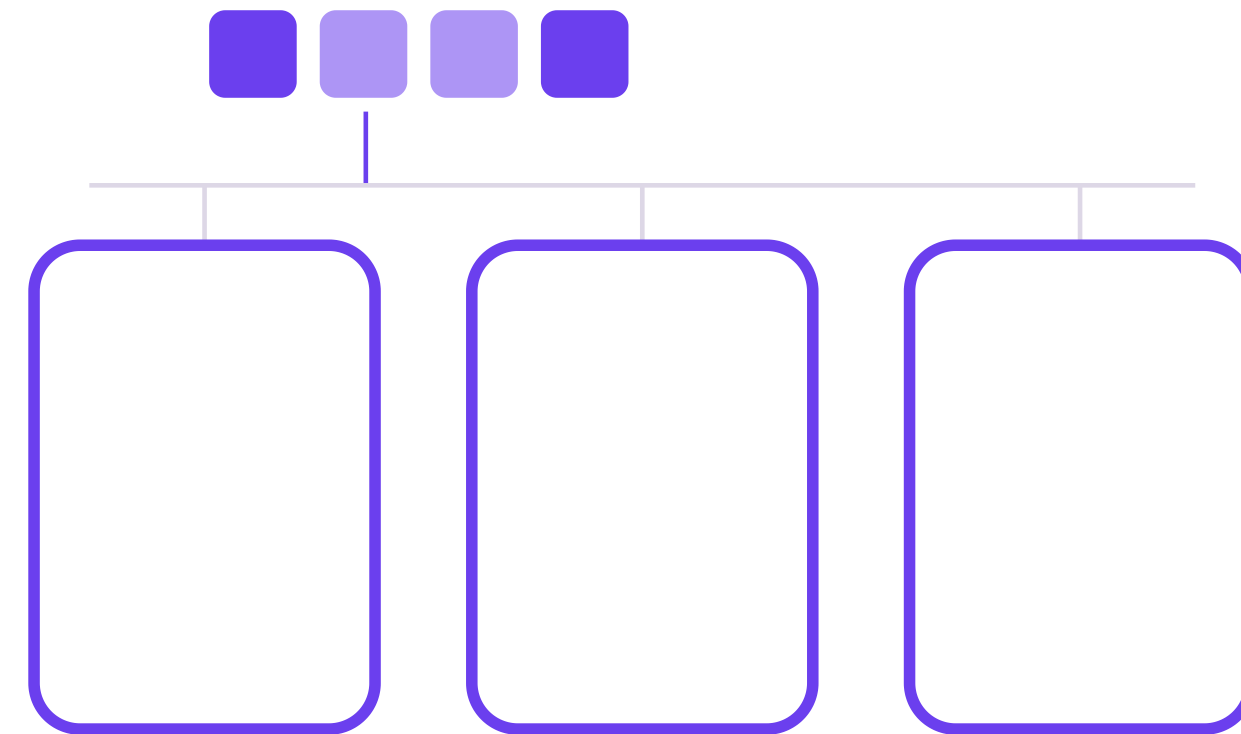
— HERE'S THE TRICK

Define the system. Screens assemble themselves.

SCREEN BY SCREEN



PATTERN FIRST



Define your components, connections, and states before any screen.

— LET'S BREAK IT DOWN

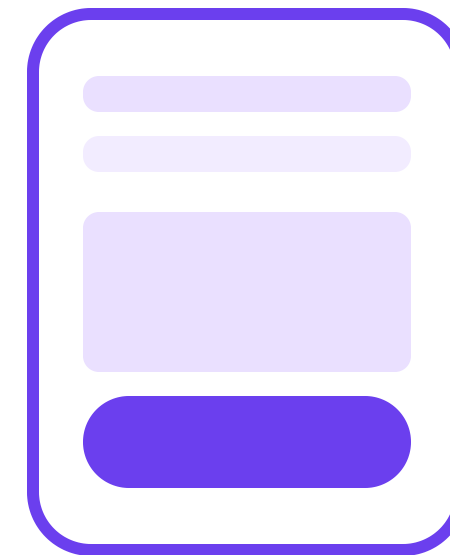
A prototype pattern is just a really clear set of instructions.

Ingredients

- Components: buttons, inputs, cards, nav
- Screens: what they are, how they connect
- States: what each component can look like



AI FOLLOWS
THE RECIPE



What pieces do you have, and what can they look like?

- ✓ **Component inventory**
Every piece + all its states
- **Interaction pattern**

Button

default · hover · loading · disabled

Input

default · filled · error · success

Card

default · hover · active · selected

Nav bar · Avatar · ...

States live inside
the component,
not a separate layer.

What triggers a change?



Component
inventory



Interaction
pattern

Triggers + conditions +
outcomes

Read

Views content, no state change
Dashboard · feed · profile

Edit

Modifies existing content
Edit profile · settings

Add

Creates something new
New post · add to cart

Confirm

Reviews and approves
Order summary · delete

Navigate

Moves between sections
Tab bar · side menu

Search

Queries or narrows a list
Results · filter · sort

Onboard

Guided through first-time setup · Welcome · tutorial · permissions

Most prototypes use 3–4 types. Label each screen before you start building.

— PUT IT ALL TOGETHER

Two layers. One document. AI builds from it.

Component inventory

states included: default · hover · loading · error · success

Interaction pattern

How screens connect, and under what conditions



Component inventory (with states inside) + Interaction pattern = your prototype pattern.

04

Build

PATTERN IN HAND. TIME TO MAKE SOMETHING REAL.

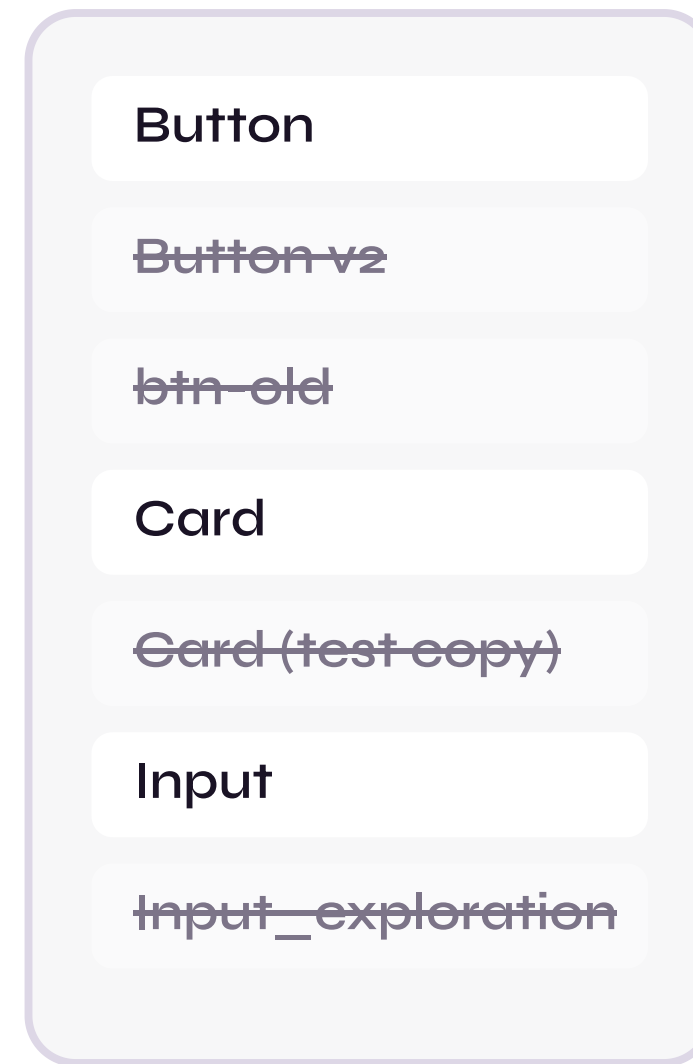
Clean up your design file **first.**

- Delete or archive components you're not using in this prototype
- Merge duplicate variants of the same component
- Remove hidden/test layers and old exploration frames
- Confirm naming is consistent across the file

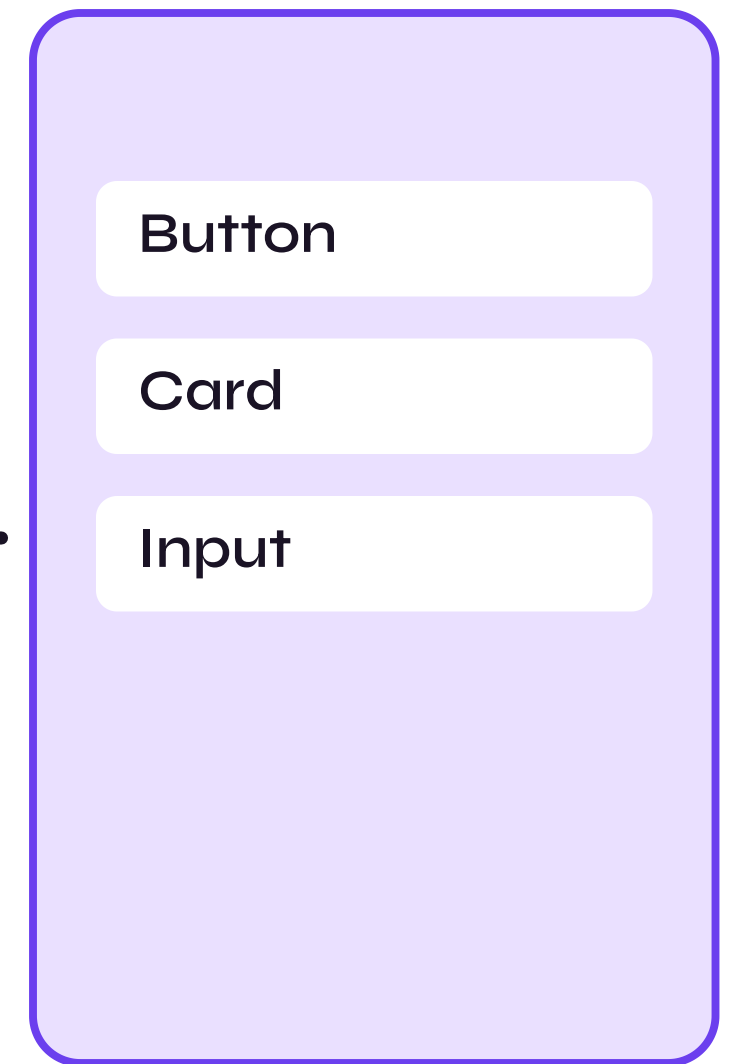
PROMPT TO USE (FOR LARGE FILES)

"Here is my Figma component list: [paste]. Which of these look like **duplicates**, **unused variants**, or **one-off experiments** rather than real system components?"

BEFORE



AFTER



Scope the build.

Don't build everything.

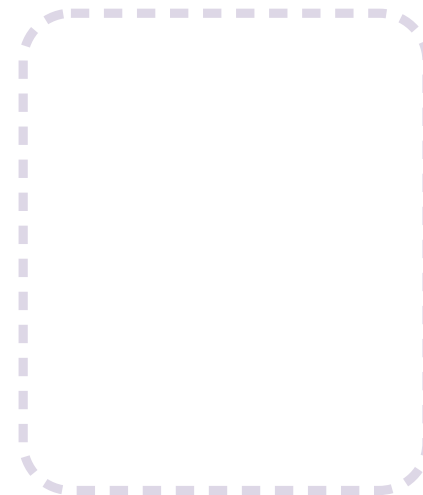
- Pick 2–3 screens from your interaction pattern.
- The goal is to prove the pattern works, not to finish the prototype.



Pick these



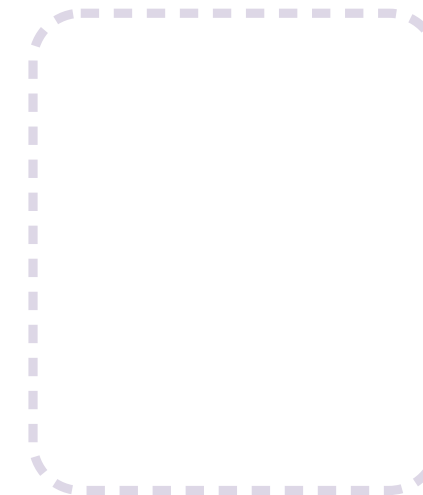
Pick these



Not yet



Not yet



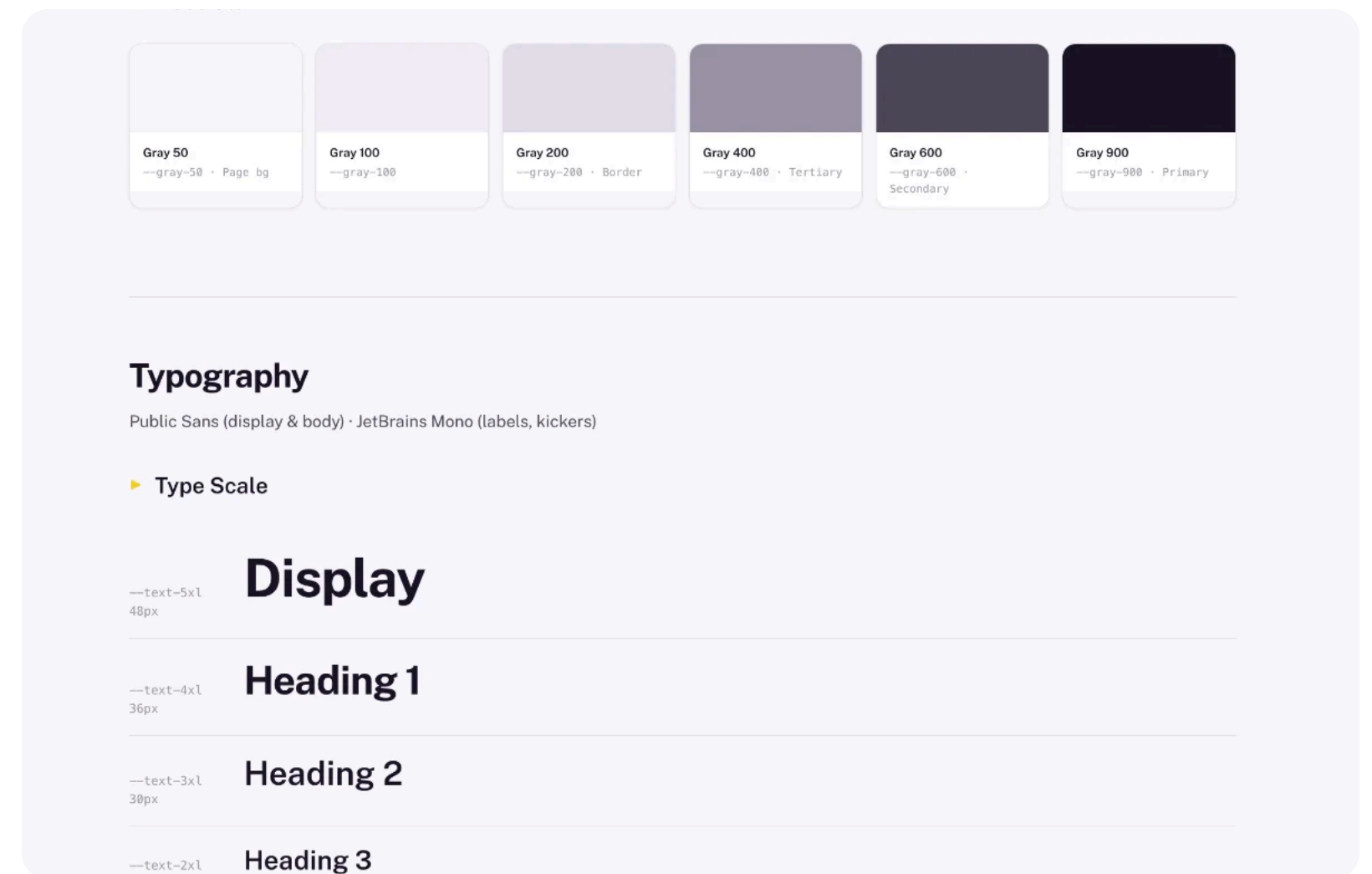
Not yet

Sync your design tokens as your design system.

PROMPT TO USE

"Read my design file via MCP. Extract the design tokens: **colour values**, **typography styles**, and **spacing**. Generate a CSS variables file from these tokens so every screen uses the exact values from my design system."

This runs once — every screen after uses your real tokens, not AI's generic defaults.

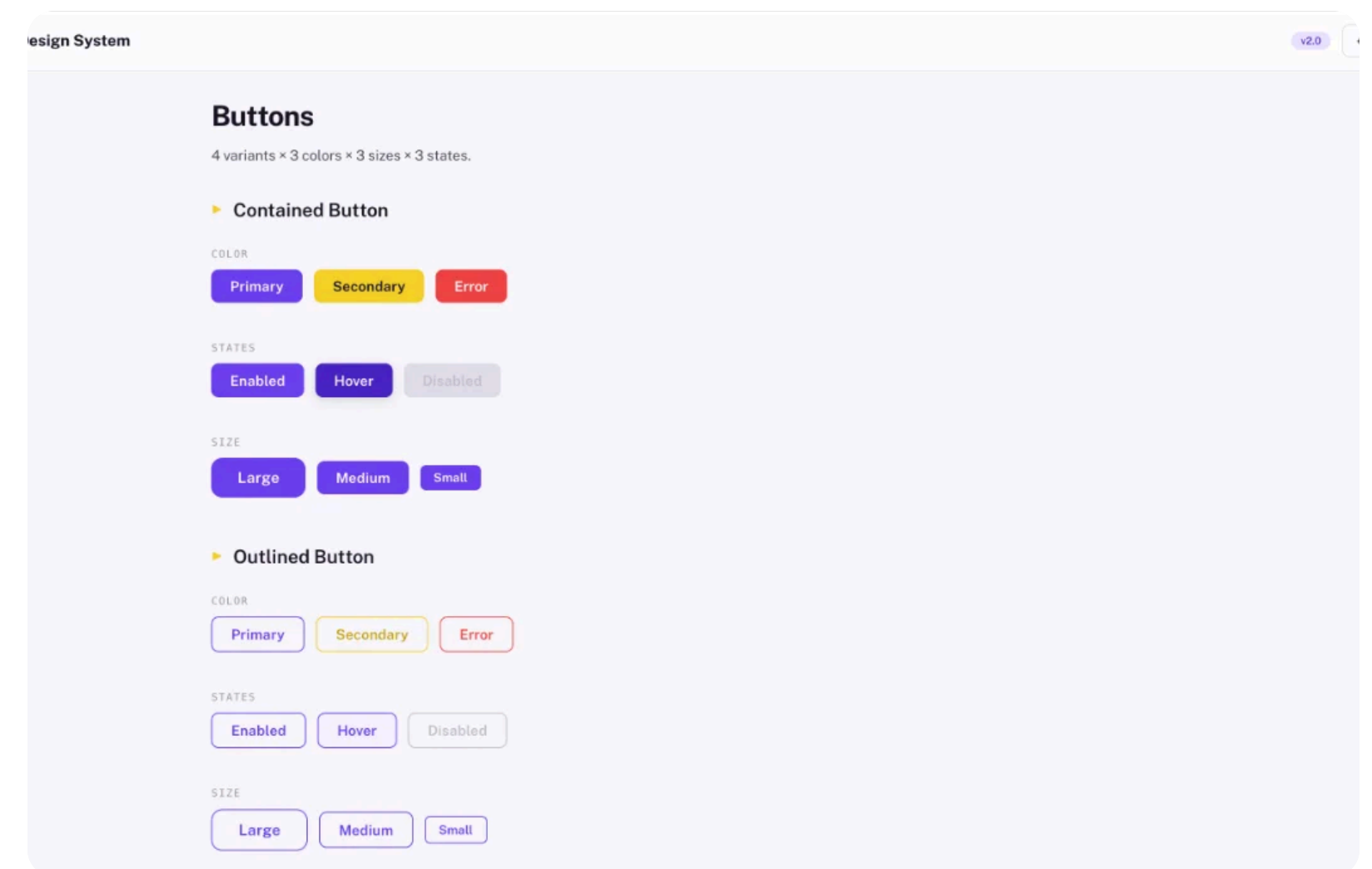


Generate the component inventory.

PROMPT TO USE

"Read my design file. Generate a component inventory: list every component with its **name**, **purpose**, and all its **states** (default, hover, loading, error, etc.). Flag anything that looks incomplete or inconsistent."

Review the output. Correct any names or states that are wrong. This becomes the source of truth for everything you build.

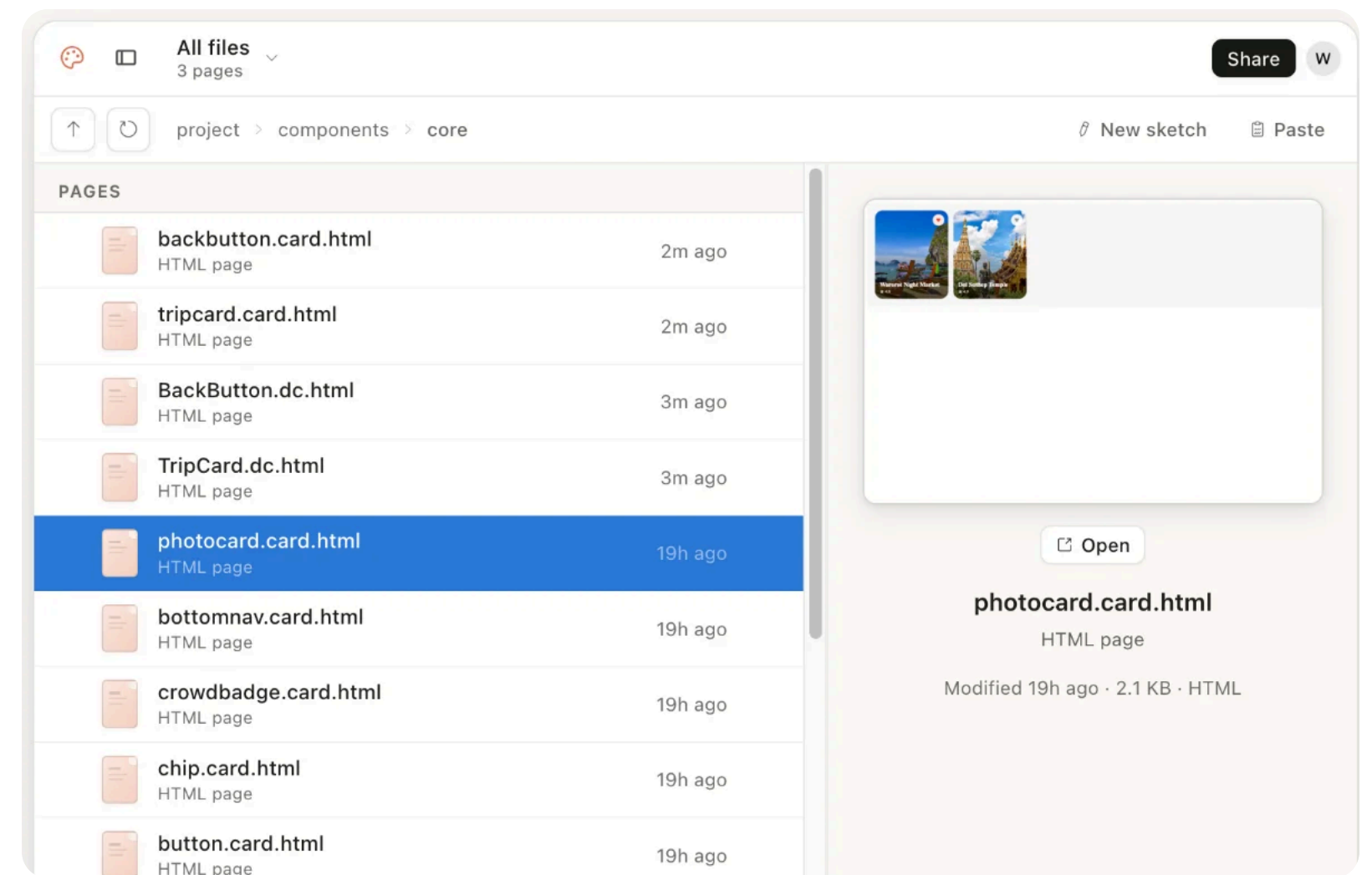


Generate the interaction pattern.

PROMPT TO USE

"Based on the component inventory, map the interaction pattern for my prototype: which **screens** exist, what **connects** them, and what **triggers** each transition. Use the component inventory as your reference."

Which screens exist, how they connect, and what triggers each move between them.

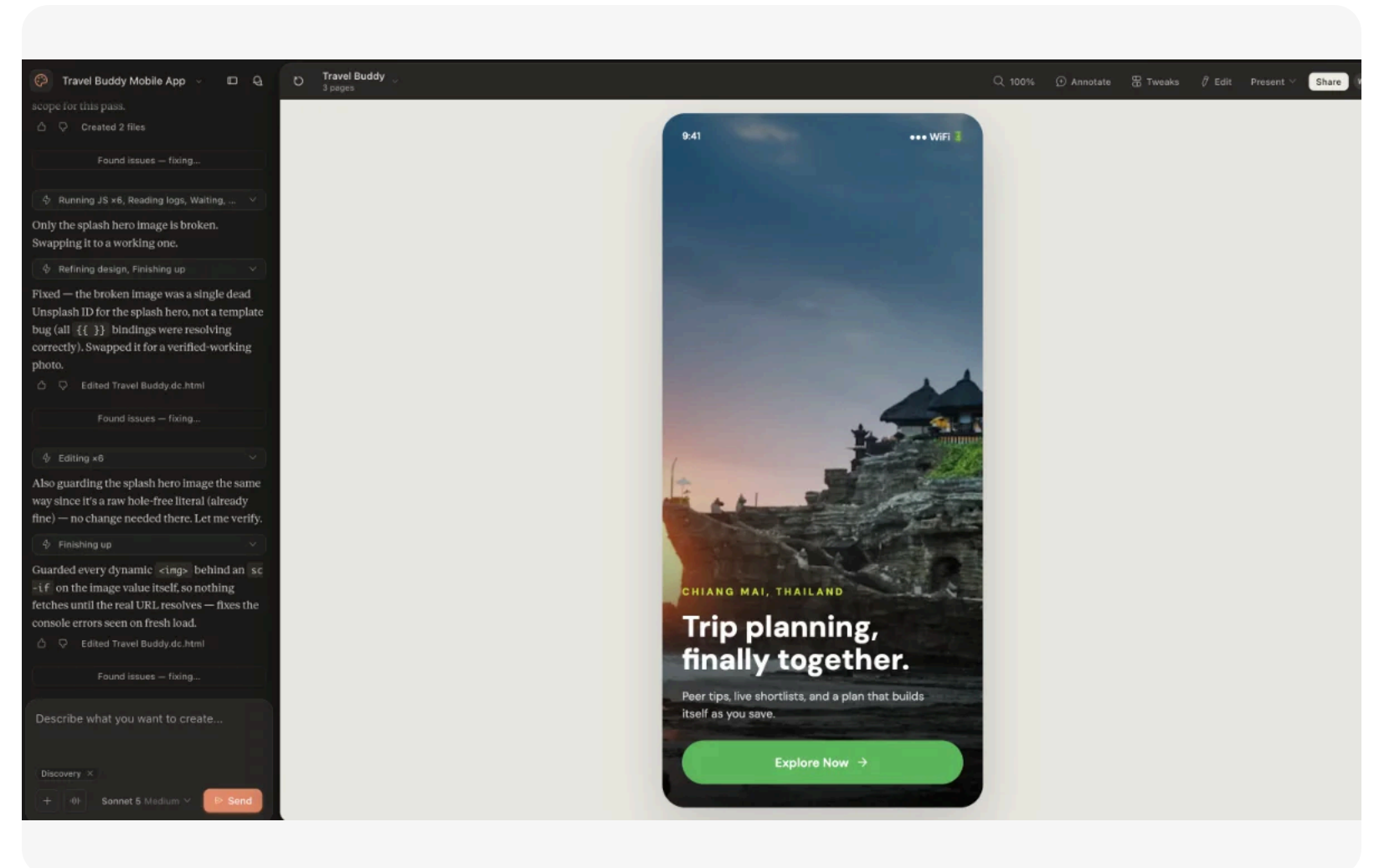


Build one screen at a time.

PROMPT TO USE

"Build the [Screen name] screen. **Goal:** [what the user is trying to do here]. **Layout:** [how things should be arranged]. **Content:** [real field names and sample data, not lorem ipsum]. **Audience:** [who's using it]. **References:** [attach screenshots or design links to match]. Use the component inventory and interaction pattern above. Match component names exactly."

Five ingredients, one complete brief: goal, layout, content, audience, references. The more screenshots or design links you attach, the closer AI matches your intent.



— AFTER EVERY BUILD

Chat. Targeted feedback. Direct edit.

CHAT

Structural changes

Use for broad changes, or anything that needs explanation.

"Rearrange this screen so the primary action is in the top row."

"Show me two alternative layouts."

TARGETED FEEDBACK

One specific element

Name exactly what's wrong and where.

"The button padding here is too tight."

"Use the primary token colour on this card."

DIRECT ADJUSTMENT

Quick nudges

Drag, resize, align directly, no prompt needed.

Fast for spacing and alignment fixes you can just place by hand.

Want a different direction without losing this version? Say so: "Save what we have and try a completely different layout."

— SET YOUR EXPECTATIONS NOW

AI builds it. **You make it right.**

01

Colours and type look generic at first

...until your token file is applied. If something looks off-brand, check that the token step ran correctly.

02

Spacing and groupings need your eye

AI places components correctly but doesn't always know what belongs together. That's design judgment, not a technical fix.

03

Each correction makes the next screen better

The feedback loop is cumulative. A screen you correct well becomes the reference for everything that follows.

ONE THING TO CARRY FORWARD

Define the system.
Generate the prototype pattern.
Check your components.
Build from what exists.
Let AI do the rest.

Five steps from design system to working prototype. The prototype pattern is the proof the system holds.

Winnie Nguyen

EXT

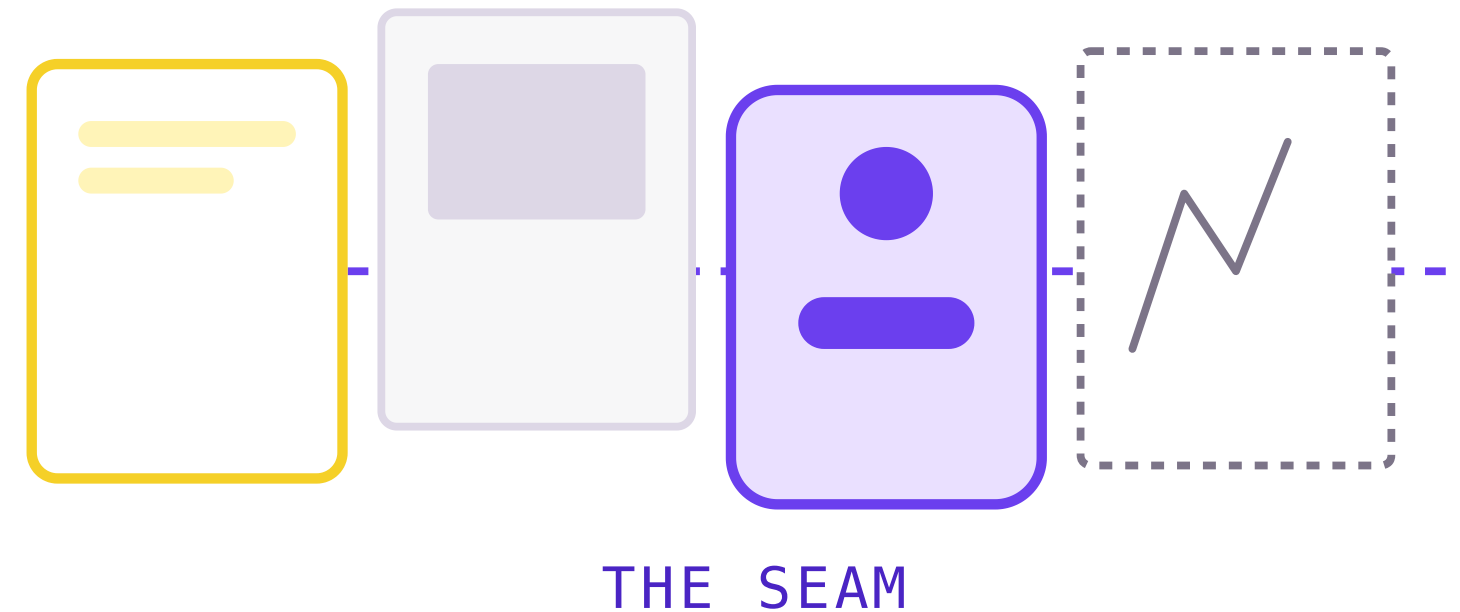
Stitching prototypes

WHEN ONE FLOW ISN'T THE WHOLE STORY

Your prototype is a flow.

The product is a journey.

"A stitch prototype is a collection of prototypes from multiple products, woven together to interactively demo a threaded, complete digital journey." Nathan Curtis, EightShapes. The parts don't need to match each other. They need to connect.



— THREE REASONS

One flow isn't the whole story.

01

Products span personas

Products span multiple flows and personas, your prototype usually doesn't. Stitching lets you prototype across personas without duplicating the design system.

02

Stakeholders see the whole

An individual demo says "here's our screen." A stitched demo says "here's what the customer experiences, start to finish." That's the conversation that surfaces real gaps.

03

Transitions hide problems

What happens between screen 2 and 3? Who triggers it? What carries over? Stitching forces you to design the seams, not just the screens.

— FIVE STEPS

Gather. Index. Connect. Demo.

1

Define the thread

One sentence: "The user's goal across all screens is [goal]"

2

Gather, don't rebuild

Figma frames, screenshots, coded screens, sketches. Missing one? Note the gap.

3

Build an index

A homepage listing every screen with its status: complete, in progress, missing

4

Add the seams

Connect each screen with a link. A clickable button is enough, don't perfect it

5

Demo the journey

Present it as a user would, start to finish. One minute, then stop.

Not a redesign. You're connecting what already exists, not rebuilding it.

Not the component. The journey.

1 · IDENTIFY MISSING SEAMS (AI CHAT TOOL)

"Here is my interaction pattern: [paste]. Here are the screens I've built: [list]. What transitions are undesigned? What does the user see between Screen 2 and Screen 3 that hasn't been prototyped?"

2 · GENERATE THE INDEX PAGE (AI CHAT TOOL)

"Here are my prototype screens: [list each with a one-sentence description and status]. Generate a simple HTML index page that lists them with status, links, and the overall journey flow."

3 · BUILD THE STITCH SHELL (AI CODING TOOL)

"I have these screens as separate HTML files: [list]. Build a navigation wrapper that stitches them into a single prototype: a homepage index, back/forward navigation, and a progress indicator."

— BEFORE YOU STITCH, KNOW THIS

It's a journey, not a deliverable.

- 01 It's a throwaway artefact.** It communicates the journey while the product is being built. The day it ships, retire it.
- 02 The goal is the journey, not the screens.** Individual screens will look imperfect. That's the point: stitching surfaces what polished demos hide.
- 03 One minute hooks the room.** Walk the user's goal, not the design decisions. If you can't tell it in a minute, the journey isn't clear enough yet.

IF YOU ONLY READ ONE THING

Stitching doesn't require consistency. **It requires connection.**

Nathan Curtis, *Stitching Prototypes* (EightShapes, 2015). The Marriott International case study is the clearest illustration of the concept. Assign as pre-reading if teaching this as a standalone session.

Winnie Nguyen